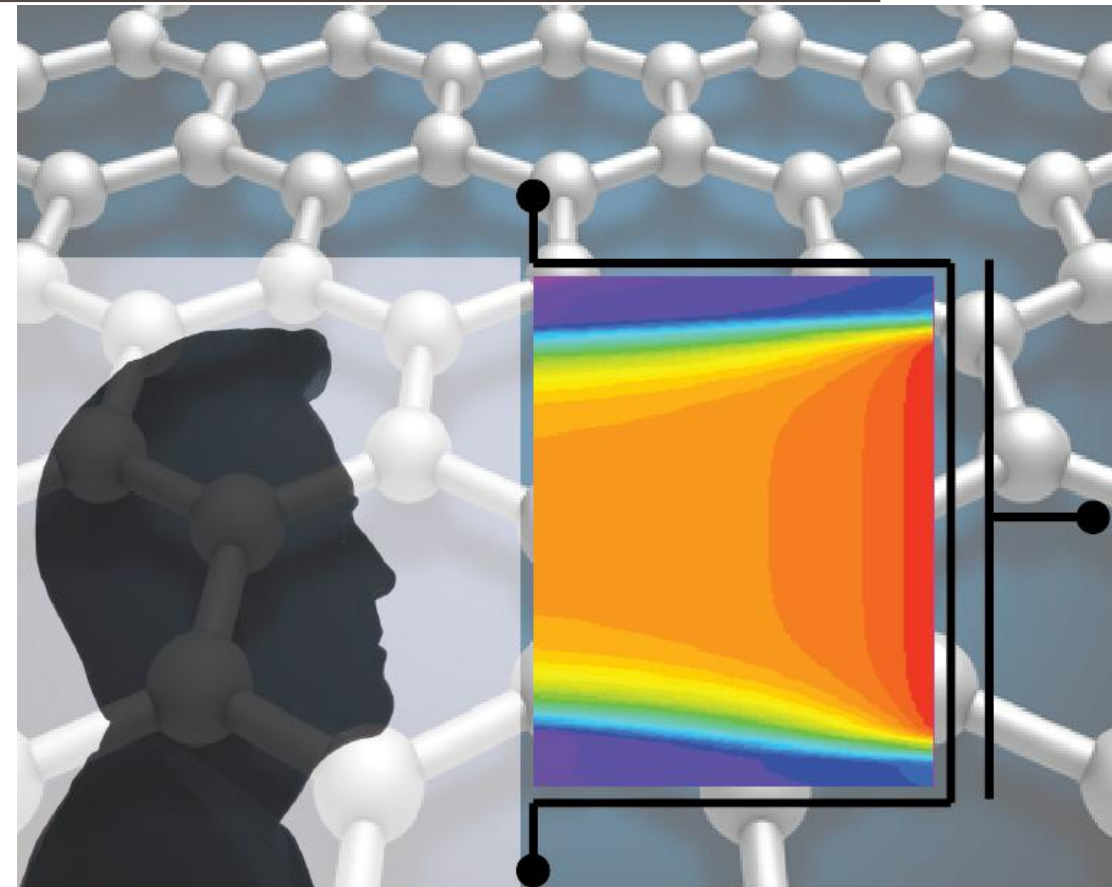


ADVANCED SOLID-STATE DEVICES

Week-8, Lecture-2

Sneh Saurabh
6th March, 2025



MOSFET



Capacitance in MOS
Structure

MOS Capacitor: Capacitance (Definition)

- Capacitance is defined as:

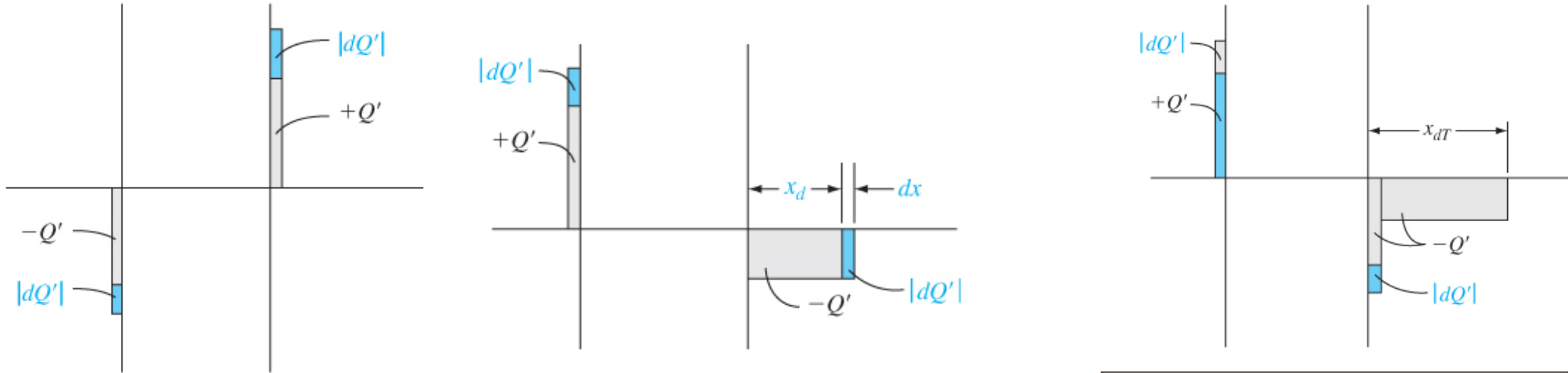
$$C = \frac{dQ}{dV}$$

dQ : magnitude of differential change in charge in one plate

dV : differential change in voltage across the capacitor

- This capacitance is a small signal capacitance or AC parameter
- Measured by superimposing a small AC voltage on an applied DC voltage
- We study the behavior of this capacitance as the DC voltage is varied

MOS Capacitor: Ideal CV (1)



Accumulation mode:

- Incremental change in voltage, leads to incremental change in charge at the two edges of oxide

$$C' = C_{ox} = \frac{\epsilon_{ox}}{t_{ox}} = \frac{\epsilon_r \epsilon_o}{t_{ox}}$$

Depletion mode:

- Incremental change in voltage, leads to incremental change in depletion layer width (and change in charge at end of depletion layer):

$$\frac{1}{C'} = \frac{1}{C_{ox}} + \frac{1}{C_d} = \frac{1}{\epsilon_{ox}/t_{ox}} + \frac{1}{\epsilon_s/x_d}$$

Inversion mode:

- Incremental change in voltage, leads to incremental change in charge at the two edges of oxide

$$C' = C_{ox} = \frac{\epsilon_{ox}}{t_{ox}} = \frac{\epsilon_r \epsilon_o}{t_{ox}}$$

MOS Capacitor: Ideal CV (p-type)

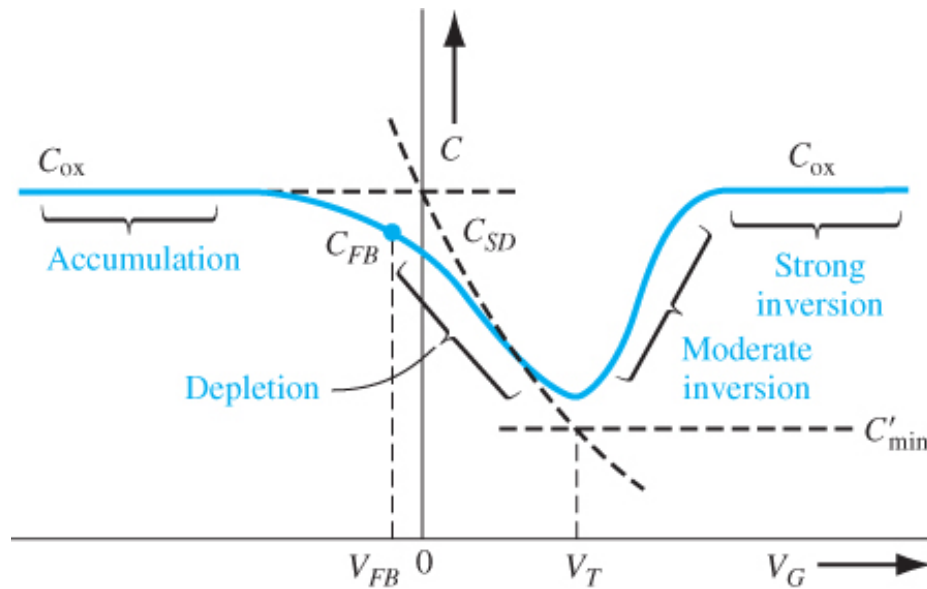


Figure 10.26 | Ideal low-frequency capacitance versus gate voltage of a MOS capacitor with a p-type substrate. Individual capacitance components are also shown.

Depletion mode: $\frac{1}{C'} = \frac{1}{\epsilon_{ox}/t_{ox}} + \frac{1}{\epsilon_s/x_d}$

$$C' = \frac{\epsilon_{ox}\epsilon_s}{t_{ox}\epsilon_s + x_d\epsilon_{ox}}$$

- As x_d increases, capacitance decreases. It reaches minimum value when depletion layer width reaches maximum.

$$C'_{min} = \frac{\epsilon_{ox}\epsilon_s}{t_{ox}\epsilon_s + x_{d,max}\epsilon_{ox}} \text{ where } x_{d,max} = \sqrt{\frac{4\epsilon_s\phi_F}{qN_a}}$$

Accumulation mode: $C' = C_{ox} = \frac{\epsilon_{ox}}{t_{ox}}$

Inversion mode: $C' = C_{ox} = \frac{\epsilon_{ox}}{t_{ox}}$

MOS Capacitor: Ideal CV (n-type)

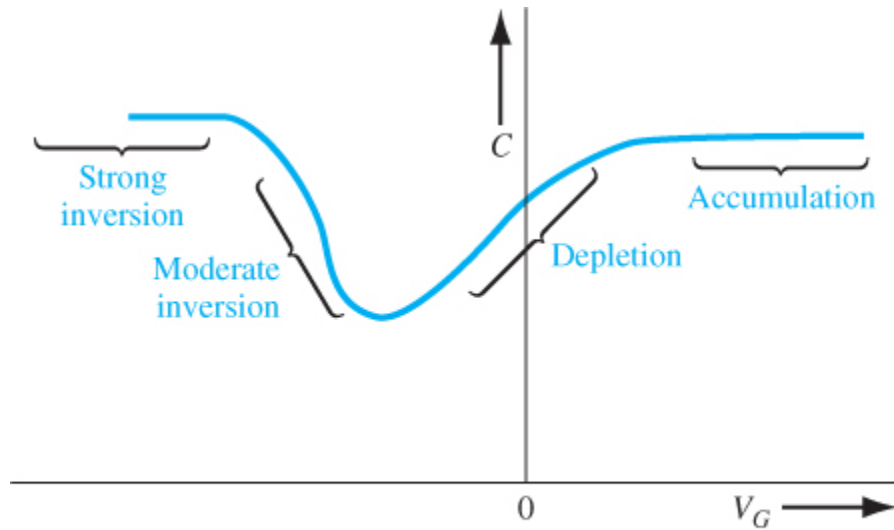
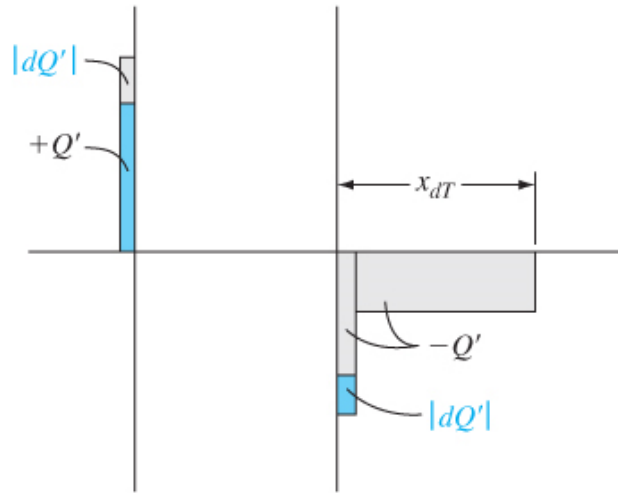


Figure 10.27 | Ideal low-frequency capacitance versus gate voltage of a MOS capacitor with an n-type substrate.

- Similar ideal CV is obtained for n-type semiconductor
- Just changing the sign of gate voltage axis

MOS Capacitor: Ideal CV at High Frequency (1)



When we make change to gate voltage, in a MOS that is in strong inversion, the charge changes in the inversion layer (at low frequency). From where does this charge (electron) come?

1. Diffusion of minority carrier from p-type substrate across the depletion region
2. Thermal generation of electron-hole pairs

At high frequency:

- Rate of change in concentration in the inversion layer may not be able to keep pace with voltage changes.
- Change in charge occurs at the end of depletion layer

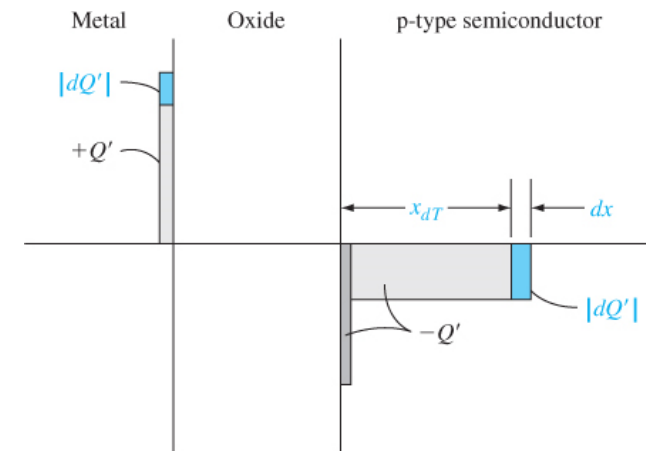


Figure 10.28 | Differential charge distribution at inversion for a high-frequency differential change in gate voltage.

MOS Capacitor: Ideal CV at High Frequency (2)

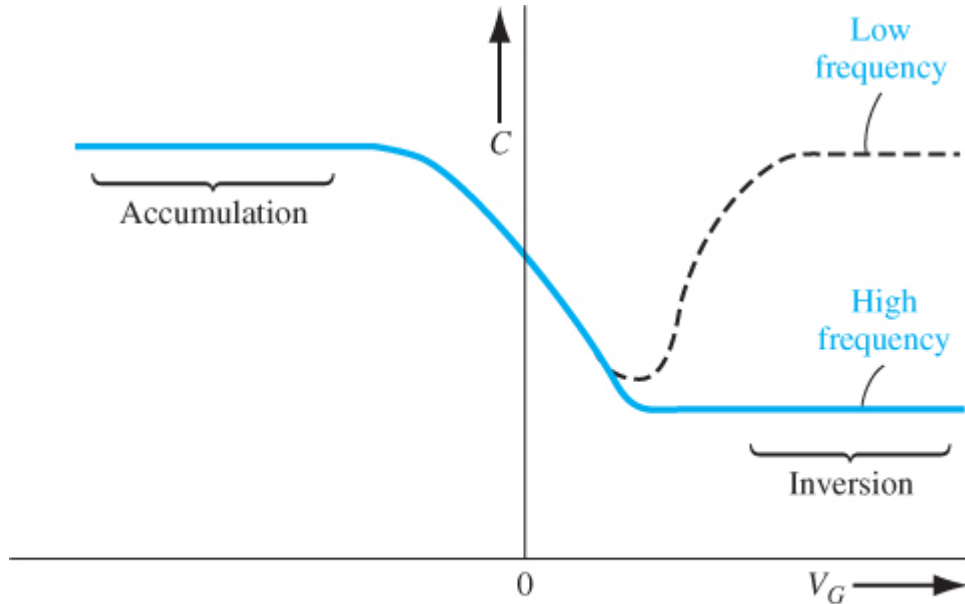


Figure 10.29 | Low-frequency and high-frequency capacitance versus gate voltage of a MOS capacitor with a p-type substrate.

- Low Frequency (5-100 Hz)

$$C' = C_{ox} = \frac{\epsilon_{ox}}{t_{ox}}$$

- High Frequency (of order of 1 MHz) or more)

$$C' = C'_{min}$$

MOS Capacitor: Effect of Fixed Oxide on CV

- Fixed oxide charge decreases the threshold of MOS with p-type Silicon
- With increasing fixed oxide charge, the CV curve move towards left

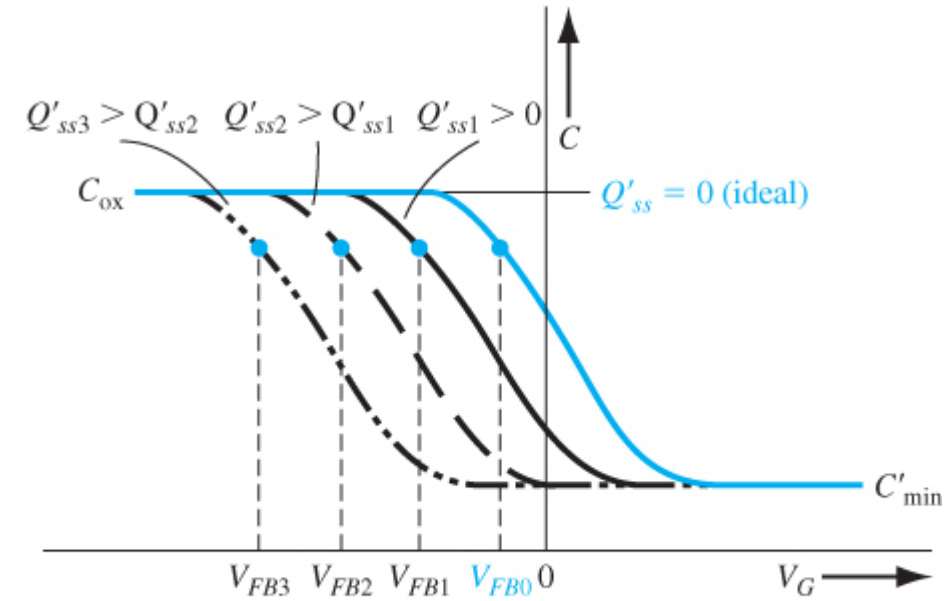


Figure 10.30 | High-frequency capacitance versus gate voltage of a MOS capacitor with a p-type substrate for several values of effective trapped oxide charge.

MOS Capacitor: Effect of Interface States on CV (1)

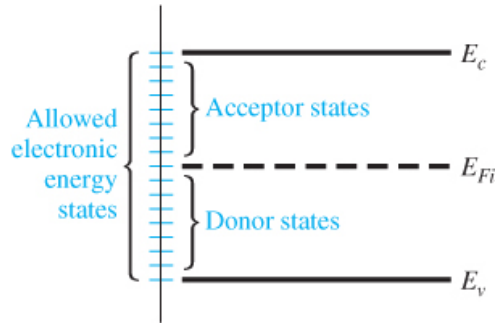


Figure 10.31 | Schematic diagram showing interface states at the oxide-semiconductor interface.

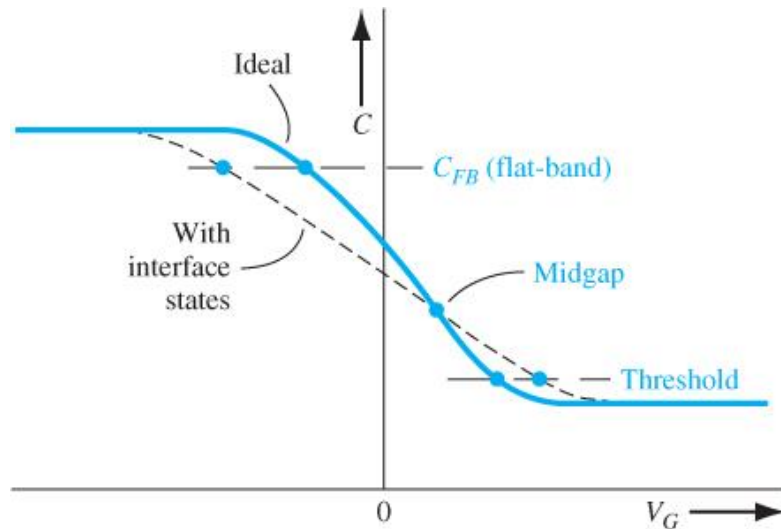


Figure 10.33 | High-frequency C-V characteristics of a MOS capacitor showing effects of interface states.

- Interface states created by dangling bonds at the surface
- Occupancy depend on the surface potential
- Charge due to interface state depends on the gate voltage
- Net charge in the interface states change from positive to negative as the MOS moves from accumulation, to depletion to inversion
- CV curve moves toward left in accumulation and toward right in inversion
- Smearing out effect due to interface trap
- CV measurement gives information about interface states

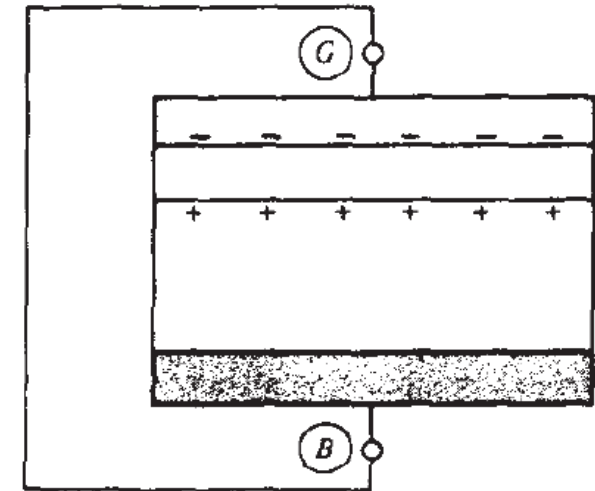
MOSFET



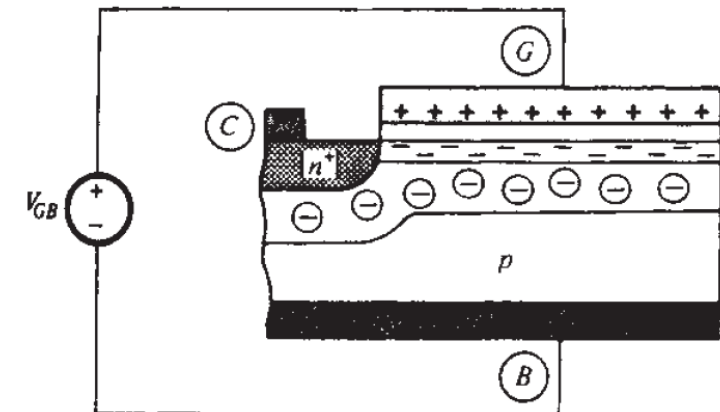
Three-Terminals

MOSFET: Three Terminal Device Abstraction

- In earlier discussions, we considered MOS capacitor
- A MOS capacitor has only **two terminals**: Gate and Body [Source and Drain are not considered for easier analysis]
- No current flows in a MOS capacitor (ideally)
- A real MOSFET is four terminal device: Gate, Body, Source, and Drain [We will now move towards real device]

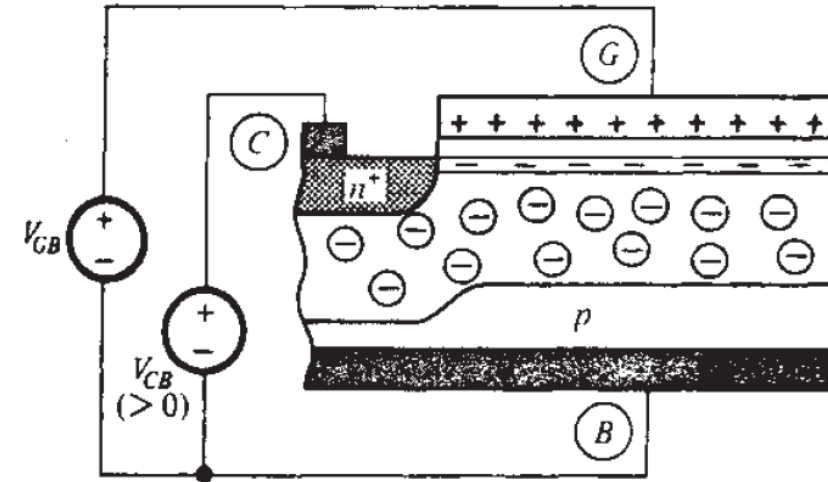


- First, we consider a three-terminal device
- Gate, Body, Terminal C (Contact)
- Terminal C is n⁺ doped (same as Source or Drain in MOSFET)
- A n⁺p junction will be formed. Regions away from C will see no effect



MOSFET: Three Terminal Device Bias

- We connect Terminal C with voltage source V_{CB}
- Need to keep $V_{CB} > 0$ so that the C-B junction is reverse-biased [other configuration is not practically useful]
- We want to study the impact of increasing V_{CB}



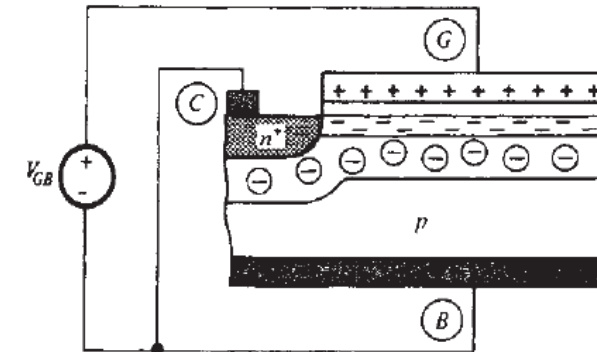
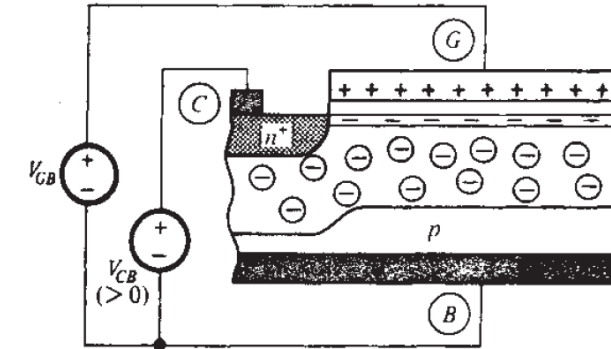
MOSFET: Impact of V_{CB} (or V_{SB}) on Inversion [1]

What will be the impact of adding the third terminal?

- A depletion region created between n^+ terminal C and the body
- The surface which was earlier isolated gets connected to the battery through the terminal C
 - Surface potential ψ_s will be impacted
 - Device is in non-equilibrium

First consider $V_{CB} = 0$

- The situation is similar to as shown in the figure
- The extra PN junction exist, which is short-circuited externally
- Will work similar to 2-terminal MOS to the right of n^+ region
- In 2-terminal MOS: $n_{surface} = N_A \exp\left(\frac{\psi_s - 2\phi_{Fp}}{kT/q}\right)$

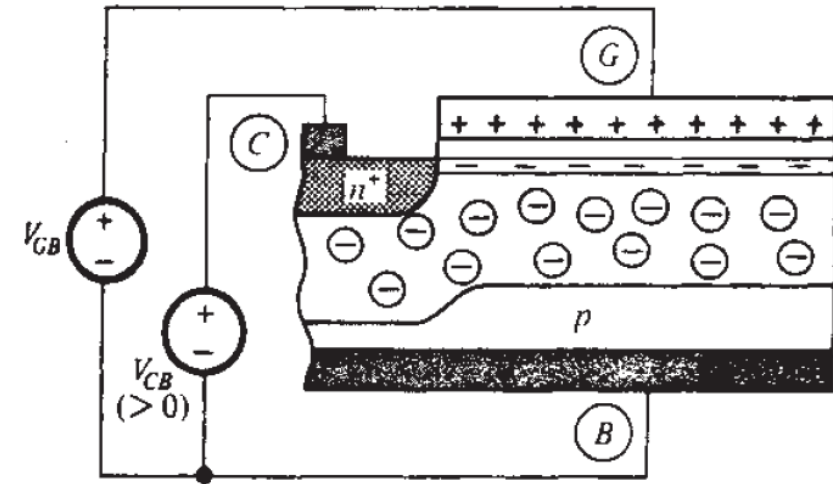


NOTE: Surface potential ψ_s “fights” against $2\phi_{Fp}$ to go into strong inversion

MOSFET: Impact of V_{CB} (or V_{SB}) on Inversion [2]

Consider $V_{CB} = 0$ and we have applied V_{GB} to obtain surface potential ψ_1 and an inversion layer is formed.

Next, consider that we have made $V_{CB} > 0$. What will be its impact on the inversion layer?

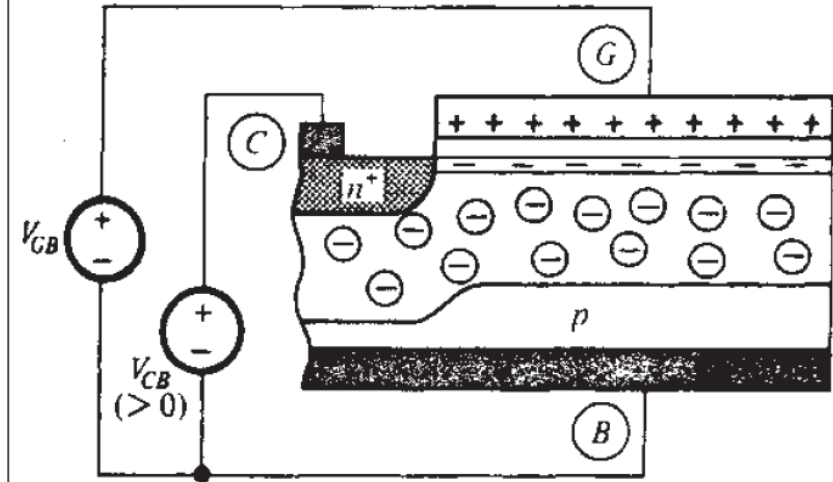


- It will attract electrons from the inversion layer and attracted electrons will flow to the battery
- The concentration of electrons in the inversion layer will decrease
- If V_{CB} is large enough, the inversion layer will disappear

MOSFET: Impact of V_{CB} (or V_{SB}) on Inversion [3]

To restore the electrons concentration to the $V_{CB} = 0$ case, what should be done to the surface potential ψ_1 ?

- The surface potential should be increased by the same amount as n+ region has increased
 - From ψ_1 to $\psi_1 + V_{CB}$
- V_{GB} needs to be increased to compensate for increased V_{CB} such that surface potential is increased from ψ_1 to $\psi_1 + V_{CB}$
- Increase in V_{GB} required will be more than V_{CB} increase



- Surface potential ψ_s “fights” against $(2\phi_{Fp} + V_{CB})$ instead of $2\phi_{Fp}$ to go into strong inversion
- Reaching strong inversion becomes more difficult

MOSFET: Impact of V_{CB} (or V_{SB}) on Threshold Voltage

Threshold Inversion Definition: Gate voltage required such that surface potential $\psi_s = 2\phi_{Fp} + V_{CB}$ instead of $\psi_s = 2\phi_{Fp}$.

Problem: Compute threshold voltage difference between the case $V_{CB} = 0$ and case $V_{CB} > 0$.

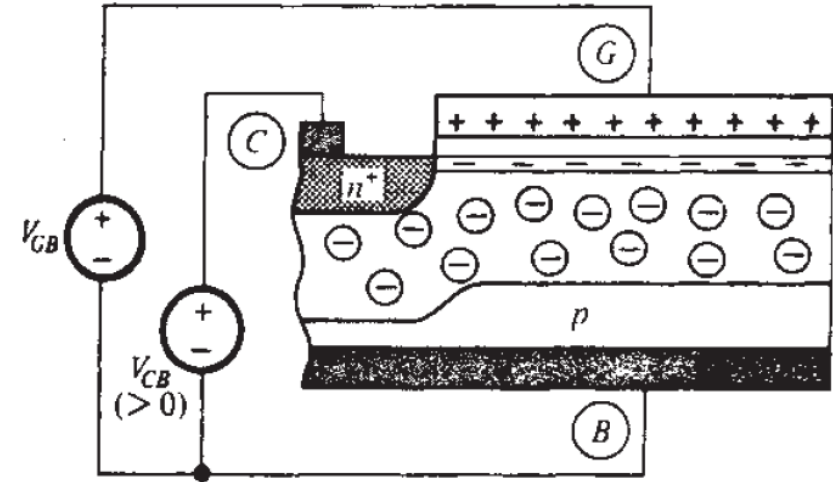
For $V_{CB} = 0$, we had computed V_{GB} required to make $\psi_s = 2\phi_{Fp}$:

$$V_T = \phi_{ms} + 2\phi_{Fp} + \frac{\sqrt{4q\epsilon_s N_a \phi_{Fp}} - Q'_{ss}}{C'_{ox}}$$

For $V_{CB} > 0$, $V_{GB} = V'_T$ make $\psi_s = 2\phi_{Fp} + V_{CB}$:

$$V'_T = \phi_{ms} + 2\phi_{Fp} + \frac{\sqrt{2q\epsilon_s N_a (2\phi_{Fp} + V_{CB})} - Q'_{ss}}{C'_{ox}}$$

$$\Delta V_T = \frac{\sqrt{2q\epsilon_s N_a (2\phi_{Fp} + V_{CB})} - \sqrt{4q\epsilon_s N_a \phi_{Fp}}}{C'_{ox}}$$



$$\Delta V_T = \frac{\sqrt{2q\epsilon_s N_a}}{C'_{ox}} \left(\sqrt{2\phi_{Fp} + V_{CB}} - \sqrt{2\phi_{Fp}} \right)$$

MOSFET: Body Effect

- Increase in threshold voltage with V_{CB} (or V_{SB} source-body voltage) is referred to as **body effect**

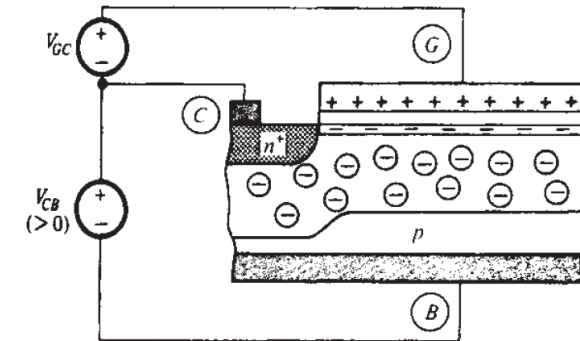
- We can write: $\Delta V_T = \frac{\sqrt{2q\epsilon_s N_a}}{C'_{ox}} (\sqrt{2\phi_{Fp} + V_{CB}} - \sqrt{2\phi_{Fp}})$ as:

$$\Delta V_T = \gamma \left(\sqrt{2\phi_{Fp} + V_{CB}} - \sqrt{2\phi_{Fp}} \right)$$

where $\gamma = \frac{\sqrt{2q\epsilon_s N_a}}{C'_{ox}}$

- We refer to γ as **body effect coefficient**

- Body effect will be more in MOSFET with:
 - Higher doping substrate doping N_a
 - Higher gate oxide thickness



C = source terminal S of MOSFET

$$V_{GC} \equiv V_{GS} \text{ and } V_{CB} \equiv V_{SB}$$

MOSFET



Current-Voltage Characteristics

Current-Voltage Characteristics: Assumptions

- We have discussed electrical behavior of four-terminal MOSFET qualitatively
- Now, we will discuss them quantitatively

Assumptions:

1. Mobility is constant and independent of electric field (bias)
 - Valid only when electric field due to drain is less than 2-3 kV/cm
 - Not valid in modern devices, will relax this assumption later
2. Gradual channel approximation: gate to substrate field is much stronger than drain to source field
 - Valid when gate length is larger than the channel depth
 - Allows us to treat device using 1-D electric field rather than 2-D electric field

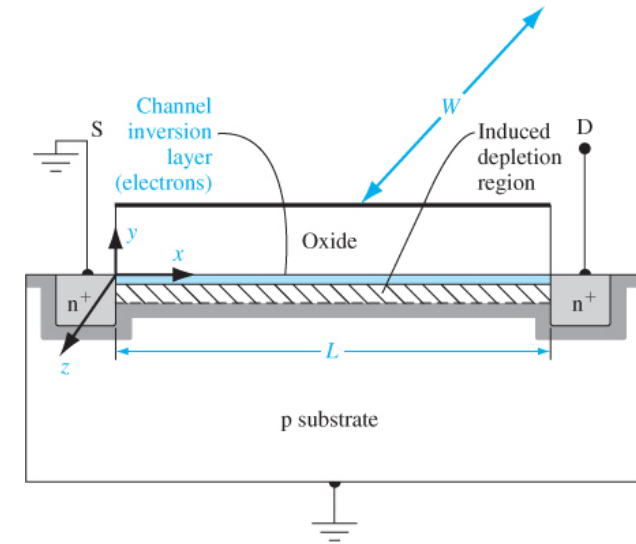


Figure 10.43 | Geometry of a MOSFET for I_D versus V_{DS} derivation.

Goal:

Find drain current in terms of drain voltage, gate voltage, and threshold voltage (assuming source is grounded) and MOSFET is in strong inversion

References

- Tsividis, Yannis. *Operation and Modeling of the MOS Transistor*. McGraw-Hill, Inc., 1987.
- D. A. Neaman, "*Semiconductor Physics and Devices*"